

Topic 05: First-Order Linear Equations & The Integrating Factor Method

Professor Leonard Master Series · Lessons 15 to 18

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Reverse Product Rule

Professor Leonard introduces the Integrating Factor method with a stroke of genius:

"Look at the Product Rule from Calculus 1: $\frac{d}{dx}[u(x)y(x)] = uy' + u'y$. Now look at a standard linear differential equation: $y' + P(x)y = Q(x)$. The left-hand side looks almost like the product rule, but the coefficients don't quite match! The Integrating Factor $\mu(x)$ is a magical mathematical catalyst. When you multiply the entire equation by $\mu(x)$, the left side instantly collapses into a single, perfect derivative!"

$$\mu(x) = e^{\int P(x)dx}$$

$$\mu(x)y' + \mu(x)P(x)y = \frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

$$\mu(x)y = \int \mu(x)Q(x)dx + C \implies y(x) = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x)dx + C \right]$$

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 5.1: Polynomial Coefficients & Initial Value Problem

Statement: Solve the IVP:

$$x \frac{dy}{dx} + 2y = 4x^2, \quad y(1) = 2$$

Step-by-Step Whiteboard Solution:

- **Step 1: Put into Standard Form (Mandatory Rule!):** Divide through by x (for $x > 0$):

$$\frac{dy}{dx} + \frac{2}{x}y = 4x$$

Identify $P(x) = \frac{2}{x}$ and $Q(x) = 4x$.

- **Step 2: Compute the Integrating Factor $\mu(x)$:**

$$\mu(x) = e^{\int P(x)dx} = e^{\int \frac{2}{x}dx} = e^{2\ln|x|} = e^{\ln(x^2)} = x^2$$

- **Step 3: Multiply the standard form by $\mu(x) = x^2$:**

$$x^2 \frac{dy}{dx} + 2xy = 4x^3$$

- **Step 4: Collapse LHS into the derivative of a product:**

$$\frac{d}{dx} [x^2 y] = 4x^3$$

- **Step 5: Integrate both sides with respect to x :**

$$x^2 y = \int 4x^3 dx = x^4 + C$$

- **Step 6: Solve for $y(x)$:**

$$y(x) = x^2 + \frac{C}{x^2}$$

- **Step 7: Apply the Initial Condition $y(1) = 2$:**

$$2 = 1^2 + \frac{C}{1^2} \implies 2 = 1 + C \implies C = 1$$

$$y(x) = x^2 + \frac{1}{x^2}$$

Problem 5.2: Exponential Integrating Factor with u -Substitution

Statement: Solve $\frac{dy}{dx} + 2xy = x$.

Step-by-Step Whiteboard Solution:

- **Step 1: Standard form check:** Already in standard form. $P(x) = 2x, Q(x) = x$.
- **Step 2: Integrating Factor:**

$$\mu(x) = e^{\int 2x dx} = e^{x^2}$$

- **Step 3: Multiply and collapse:**

$$\frac{d}{dx} [e^{x^2} y] = x e^{x^2}$$

- **Step 4: Integrate RHS:**

$$e^{x^2} y = \int x e^{x^2} dx$$

Let $u = x^2$, $du = 2x dx \implies x dx = \frac{1}{2} du$.

$$e^{x^2} y = \frac{1}{2} e^{x^2} + C$$

- **Step 5: Divide by e^{x^2} :**

$$y(x) = \frac{1}{2} + C e^{-x^2}$$

Notice that as $x \rightarrow \pm\infty$, $y(x) \rightarrow \frac{1}{2}$ (the steady-state equilibrium!).

Problem 5.3: Trigonometric Integrating Factor

Statement: Solve $(\cos x) \frac{dy}{dx} + (\sin x)y = 1$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Step-by-Step Whiteboard Solution:

- **Step 1: Standard Form:** Divide by $\cos x$:

$$\frac{dy}{dx} + (\tan x)y = \sec x$$

- **Step 2: Integrating Factor:**

$$\mu(x) = e^{\int \tan x dx} = e^{\ln |\sec x|} = \sec x$$

- **Step 3: Multiply and collapse:**

$$\frac{d}{dx} [(\sec x)y] = \sec^2 x$$

- **Step 4: Integrate both sides:**

$$(\sec x)y = \int \sec^2 x dx = \tan x + C$$

- **Step 5: Solve for $y(x)$:**

$$y(x) = \frac{\tan x + C}{\sec x} = \sin x + C \cos x$$

Problem 5.4: Integrating Factor with Integration by Parts

Statement: Solve $x \frac{dy}{dx} - y = x^2 \sin x$.

Step-by-Step Whiteboard Solution:

- **Step 1: Standard Form:**

$$\frac{dy}{dx} - \frac{1}{x}y = x \sin x$$

- **Step 2: Integrating Factor (Watch the Negative Sign!):**

$$\mu(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = e^{\ln(x^{-1})} = x^{-1} = \frac{1}{x}$$

- **Step 3: Multiply and collapse:**

$$\frac{d}{dx} \left[\frac{y}{x} \right] = \frac{1}{x} (x \sin x) = \sin x$$

- **Step 4: Integrate:**

$$\frac{y}{x} = \int \sin x dx = -\cos x + C$$

- **Step 5: Multiply by x :**

$$y(x) = -x \cos x + Cx$$

Problem 5.5: Inverting Variables (x as a Function of y)

Statement: Solve $\frac{dy}{dx} = \frac{1}{x+y^2}$.

Step-by-Step Whiteboard Solution:

- **Step 1: Recognize Non-Linearity in y :** dy/dx has y^2 in the denominator. It is impossible to separate or make linear in y .
- **Step 2: Flip the derivative:**

$$\frac{dx}{dy} = x + y^2 \implies \frac{dx}{dy} - x = y^2$$

This is a **1st-order linear ODE for x as a function of y !**

- **Step 3: Integrating Factor in terms of y :**

$$\mu(y) = e^{\int -1 dy} = e^{-y}$$

- **Step 4: Multiply and collapse:**

$$\frac{d}{dy} [e^{-y} x] = y^2 e^{-y}$$

- **Step 5: Integrate RHS using tabular integration by parts:**

$$\int y^2 e^{-y} dy = -y^2 e^{-y} - 2y e^{-y} - 2e^{-y} + C$$

$$e^{-y} x = -e^{-y} (y^2 + 2y + 2) + C$$

- **Step 6: Multiply by e^y :**

$$x(y) = -(y^2 + 2y + 2) + C e^y$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting to Divide by Leading Coefficient:** Finding $\mu(x)$ from the non-standard form is the #1 reason students fail this problem on exams!
- **Trap 2: Losing the Minus Sign:** $\int -\frac{2}{x} dx = -2 \ln x = \ln(x^{-2})$. Do NOT write x^2 !
- **Trap 3: Forgetting to Divide $+C$:** Writing $y = \int \mu Q dx + C$ instead of $y = \frac{1}{\mu} [\int \mu Q dx + C]$. $+C$ must be divided by $\mu(x)$!